Software Engineering

PRISM
Formal Modeling and Analysis

- Exercises -

Exercise 1

- Source: Dynamic Power Management (DPM) case study available from

http://www.prismmodelchecker.org/courses/pmc1112/
http://www.prismmodelchecker.org/tutorial/

- Start from (P, S and R) properties given in the course lecture slides

1. Add a reward structure to the model called "lost", which assigns 1 to every transition of the model labelled with action “request” from a state where the queue is full. Next, create a new property to check the expected cumulated reward up until time T. Don't forget to specify which reward structure you want in the property.
2. The performance of the system is actually a trade-off between several factors. In the sequel we study the power consumption of the system, asumming the energy is used at rate 0.13, 0.95 or 2.15 for SP power states "sleep","idle" and "busy", respectively. In addition, transitions between power states have a fixed energy cost: going from "sleep" to "idle" uses 7.0 units and from "idle" to "sleep" uses 0.067 units. Add a reward structure to the model representing the power consumption. Use a cumulative reward property to investigate the energy consumption over time of the system.

Exercise 2

The following PRISM code describes a DTMC:

```
dtmc
module M1
  v1 : [0..1] init 0;
  []  v1=0 & v2=0 -> 0.9:(v1'=0) + 0.1:(v1'=1);
  [a] v1=0 & v2=1 -> 1:(v1'=1);
  [b] v1=1 -> 1:true;
endmodule
module M2
  v2 : [0..1] init 0;
  []  v1=0 & v2=0 -> 0.7:(v2'=0) + 0.3:(v2'=1);
  [a] v1=0 & v2=1 -> 1:true;
  [b] v1=1 -> 1:true;
endmodule
```

[1] The PRISM language semantics, available from: http://www.prismmodelchecker.org/doc/semantics.pdf

Define the semantics of this PRISM model following [1]

a) Construct the system module from the component modules

b) Give the semantics of the system module as a transition probability matrix $P:S\times S\rightarrow[0,1]$

- We recall that a (discrete) probability distribution over a countable set S is a function $\mu:S\rightarrow[0,1]$ satisfying $\sum_{s\in S} \mu(s)=1$.
- We use $[s_0 \rightarrow p_0,\ldots,s_n \rightarrow p_n]$ to denote the distribution that chooses s_i with probability p_i (for $1\leq i\leq n$) and Dist(S) for the set of distributions over S

Exercise 3

The following PRISM code describes a CTMC:

```
ctmc
module M1
v1 : [0..1] init 0;
[] v1=0 & v2=0 -> 4.5:(v1'=0) + 0.5:(v1'=1);
[a] v1=0 & v2=1 -> 1:(v1'=1);
[b] v1=1 -> 1:true;
endmodule
module M2
v2 : [0..1] init 0;
[] v1=0 & v2=0 -> 3.5:(v2'=0) + 1.5:(v2'=1);
[a] v1=0 & v2=1 -> 1:true;
[b] v1=1 -> 2:true;
endmodule
```

[1] The PRISM language semantics, available from: http://www.prismmodelchecker.org/doc/semantics.pdf

Define the semantics of this PRISM model following [1]

a) Construct the system module from the component modules
b) Give the semantics of the system module as a transition rate matrix $R: S \times S \rightarrow \mathbb{R}_{\geq 0}$

Exercise 4

The following PRISM code describes a MDP:

```
mdp
module M1
   v1 : [0..1] init 0;
   []  v1=0 & v2=0 -> 0.9:(v1'=0) + 0.1:(v1'=1);
   [a] v1=0 & v2=1 -> 1:(v1'=1);
   [b] v1=1 -> 1:true;
endmodule
module M2
   v2 : [0..1] init 0;
   []  v1=0 & v2=0 -> 0.7:(v2'=0) + 0.3:(v2'=1);
   [a] v1=0 & v2=1 -> 1:true;
   [b] v1=1 -> 1:true;
endmodule
```

[1] The PRISM language semantics, available from: http://www.prismmodelchecker.org/doc/semantics.pdf

Define the semantics of this PRISM model following [1]

a) Construct the system module from the component modules

b) Give the semantics of the system module as the function (given in [1]):

$$\mathrm{Steps} : S \rightarrow 2^{\mathrm{Dist}(S)}$$

- S is the set of states, $2^{\mathrm{Dist}(S)}$ =power set of Dist(S), i.e., the set of all subsets of Dist(S)

c) Give the semantics of the system module when $\mathrm{Steps}: S \rightarrow 2^{(\mathrm{Act} \cup \{\uparrow\}) \times \mathrm{Dist}(S)}$ is (re)defined as follows:

$$\mathrm{Steps}(s) = \{(\alpha , \mu_{c,s}) \mid c \in C,\ s \in S\}$$

- $(\alpha \in)$ Act = set of action labels, $\uparrow \not\in$ Act,
- α=a, if $c = [a]\ g \rightarrow \lambda_1{:}u_1 + \ldots + \lambda_n{:}u_n$
- $\alpha=\uparrow$, if $c = []\ g \rightarrow \lambda_1{:}u_1 + \ldots + \lambda_n{:}u_n$
- $\mu_{c,s}$: Dist(S) - $\mu_{c,s}$ are given in [1]